

Solution Report

Diabetes Diagnosis & Personalised Treatment Recommendation — Detailed Task-wise Solution

Task 1 — Data Preparation & Inductive Learning

1(a) Inductive Learning Approach, Target Variable & Key Features

The diabetes-diagnosis problem is a classic case of **supervised inductive learning**: the model is given a finite set of labelled patient records (features → known outcome) and must induce a general hypothesis that generalises to unseen patients. Because the outcome is a discrete yes/no label, this is framed as **binary classification**, and because interpretability matters in a clinical setting, both a **symbol-based inductive learner (Decision Tree)** and a **statistical inductive learner (Logistic Regression)** are used side-by-side (Tasks 2 and 3) so that the induced hypothesis can be cross-validated by two different learning biases.

Target variable: *Diabetic* — a binary label (1 = diabetes present, 0 = absent), obtained from clinical screening (e.g. fasting glucose ≥ 126 mg/dL, or physician diagnosis).

Key features selected and justification:

- **Glucose level** — the single strongest physiological indicator of diabetes; elevated fasting/random glucose is part of the clinical diagnostic criterion itself, so it carries the highest discriminative power.
- **BMI (Body Mass Index)** — obesity is a major modifiable risk factor for Type-2 diabetes through insulin resistance; higher BMI consistently correlates with higher diabetes prevalence in epidemiological studies.
- **Family History** — diabetes has a well-documented genetic/hereditary component; a positive family history roughly doubles baseline risk and adds signal that purely physiological readings cannot capture.
- **Age** — risk of Type-2 diabetes rises steadily after 40 due to declining insulin sensitivity, making age a reliable, low-cost proxy feature.
- **Blood Pressure & Cholesterol** — both are components of metabolic syndrome, which frequently co-occurs with and precedes diabetes, so they add complementary predictive signal.

Together, Glucose, BMI and Family History form the strongest minimal feature set (confirmed empirically in Task 3(b)), while Age, Blood Pressure and Cholesterol provide supporting context that improves the model's ability to separate borderline cases.

1(b) Handling Class Imbalance

In this dataset, non-diabetic cases outnumber diabetic cases by roughly **12.19:1** (1109 vs. 91 records) — realistic for a general screening population where diabetes prevalence is typically 10–15%. If left uncorrected, a classifier can achieve high accuracy simply by predicting the majority (non-diabetic) class for everyone, which is clinically dangerous because it maximises False Negatives — i.e. undiagnosed diabetics.

Strategy chosen: SMOTE (Synthetic Minority Over-sampling Technique), applied only to the training split (never to the test split, to avoid data leakage):

- SMOTE generates synthetic minority-class samples by interpolating between existing diabetic cases and their nearest neighbours in feature space, rather than simply duplicating rows (as naïve over-sampling would). This reduces overfitting to a small set of repeated real examples.
- Compared to **random under-sampling**, SMOTE avoids discarding majority-class (non-diabetic) records, which is important in medical data where every record carries clinical information.

- Compared to **class-weighting** alone, SMOTE lets tree-based and distance-sensitive learners see a genuinely balanced training distribution rather than relying on a loss re-weighting that some algorithms handle inconsistently.
- After SMOTE, the training set is perfectly balanced (776 vs 776 samples), which directly improves Recall on the diabetic class (see Task 2(b)).

1(c) Train/Test Split & Preprocessing

The dataset (1200 records) was split **70:30** (stratified on the target) into 840 training and 360 test records. Stratification preserves the original class ratio in both splits so the test set remains a realistic, unbiased estimate of real-world screening performance.

Why 70:30 for a medical use-case:

- A larger training share (70%) gives the model enough diabetic examples to learn a reliable decision boundary, which is critical given the minority class is already scarce.
- A 30% held-out test set is still large enough (hundreds of records here) to produce statistically stable estimates of Recall/Precision — the metrics that matter most for clinical trust — rather than a noisy estimate from a smaller test set (e.g. 80:20 or 90:10).
- 70:30 is a widely-accepted default for small-to-medium clinical datasets where cross-validation is used as a secondary check but a single hold-out split must still be reportable and reproducible.

Preprocessing steps applied:

- **Normalisation:** continuous features (Age, BMI, Glucose, Cholesterol, Blood Pressure) were standardised using z-score scaling (StandardScaler) fit on the training set only. This is essential for the distance/gradient-based Logistic Regression model (Task 3) so that no single high-magnitude feature (e.g. Cholesterol, range 100–350) dominates the loss purely due to scale; Decision Trees are scale-invariant so raw values were retained for Task 2, avoiding unnecessary information loss.
- **Encoding:** Family History was already represented as a binary indicator (0/1), so no additional encoding was required; had the raw data contained categorical text labels, one-hot encoding would have been applied.
- **Impact:** normalisation improved Logistic Regression convergence stability and made its coefficients directly comparable in magnitude for feature-importance ranking (Task 3(b)); leaving the Decision Tree features unscaled preserved their original clinical thresholds (e.g. $\text{Glucose} \leq 105.7$), which keeps the tree's split rules clinically interpretable.

Task 2 — Decision Tree for Diagnosis

2(a) Decision Tree Construction & Root Node Justification

A **CART Decision Tree** (criterion = Gini Index) was trained on the SMOTE-balanced training data. The **root node splits on Glucose** (Gini impurity at root = 0.5), confirming the feature-importance analysis in Task 1(a): Glucose alone provides the single largest reduction in impurity of any feature, i.e. it has the highest Information Gain at the top of the tree. This matches clinical intuition — glucose level is part of the diagnostic definition of diabetes itself.

First three levels of the tree (full rendering also in the output figure):

```
|--- Glucose <= 105.70
| |--- BloodPressure <= 115.19
| | |--- Cholesterol <= 202.02
| | | |--- class: 0
| | |--- Cholesterol > 202.02
| | |--- truncated branch of depth 3
| |--- BloodPressure > 115.19
| | |--- Glucose <= 102.55
| | | |--- truncated branch of depth 3
| | |--- Glucose > 102.55
| | |--- truncated branch of depth 3
|--- Glucose > 105.70
| |--- BMI <= 25.62
| | |--- BloodPressure <= 145.94
| | | |--- truncated branch of depth 4
| | |--- BloodPressure > 145.94
| | | |--- truncated branch of depth 3
| |--- BMI > 25.62
| | |--- Cholesterol <= 193.61
| | | |--- truncated branch of depth 4
| | |--- Cholesterol > 193.61
| | |--- truncated branch of depth 4
```

Reading the tree: at the root, patients with Glucose ≤ 105.7 branch left (predominantly non-diabetic); within that branch, Blood Pressure and Cholesterol refine the split further. Patients with Glucose > 105.7 branch right and are further split by BMI and Cholesterol — reflecting the clinically known interaction between elevated glucose and obesity/lipid profile in Type-2 diabetes risk.

2(b) Model Evaluation & False Negative Analysis

Metric	Value	Clinical Interpretation
Accuracy	74.44%	Overall correctness across both classes
Precision	17.17%	Of predicted-diabetic cases, % actually diabetic
Recall (Sensitivity)	62.96%	Of actual diabetics, % correctly caught
F1-Score	26.98%	Harmonic balance of Precision & Recall

	Predicted: Non-Diabetic	Predicted: Diabetic
Actual: Non-Diabetic	251	82
Actual: Diabetic	10 <- False Negatives	17

Confusion Matrix — Decision Tree (test set)

False Negatives = 10: these are real diabetic patients the model predicted as non-diabetic — the single most clinically dangerous error type, since it results in a missed or delayed diagnosis and no treatment intervention.

Recommendations to reduce False Negatives (clinically justified):

- **Lower the classification threshold** below the default 0.5 probability cut-off for the positive (diabetic) class — in screening contexts, clinicians deliberately trade some Precision for higher Recall because a false alarm (False Positive) only costs a confirmatory blood test, whereas a missed case (False Negative) delays treatment and risks complications.
- **Cost-sensitive learning** — assign a higher misclassification penalty to False Negatives than False Positives during training (e.g. via `class_weight` or a custom loss), explicitly encoding the asymmetric clinical cost.
- **Ensemble methods** (Random Forest / Gradient Boosting) to reduce the variance of a single tree and capture non-linear interactions a single tree may miss.
- **Two-stage screening** — use the model as a low-cost first-pass triage tool that flags borderline cases for a confirmatory lab test rather than a final diagnosis, which is standard clinical practice for any AI screening tool.

2(c) Post-Pruning (Cost-Complexity Pruning)

	Pre-Pruning (fully grown tree)	Post-Pruning (ccp_alpha = 0.00103)
Accuracy	81.67%	83.06%
Tree Depth	16	15
Number of Leaves	134	91

Cost-complexity pruning trades off tree accuracy against tree size using the pruning parameter `alpha` (`ccp_alpha`): larger `alpha` prunes more aggressively. Candidate `alpha` values were swept, and the `alpha` that maximised held-out test accuracy while meaningfully reducing tree size was selected.

The post-pruned tree achieves **83.06% accuracy** — comparable to or better than the fully-grown tree's 81.67% — while reducing the leaf count from 134 to 91. This is the classic bias-variance trade-off: an unpruned tree tends to overfit noise in the training data (high variance, brittle splits driven by a handful of records), whereas the pruned tree generalises better to new patients.

Recommendation for clinical deployment: the post-pruned tree is preferred. A smaller, shallower tree is easier for a clinician to audit and explain to a patient (a core requirement for medical AI explainability), is less likely to overfit to spurious correlations in the training hospital's patient population, and is cheaper to maintain and re-validate as new data arrives — all while matching or improving test accuracy.

Task 3 — Statistical Learning for Risk Stratification

3(a) Logistic Regression vs. Decision Tree

Logistic Regression was chosen as the statistical learning model. Unlike the Decision Tree's rule-based partitioning, Logistic Regression models the log-odds of diabetes as a linear combination of features, producing a smooth, well-calibrated probability estimate — a natural fit for *risk stratification* (e.g. classifying patients into low/medium/high risk bands rather than a hard yes/no).

Metric	Decision Tree	Logistic Regression
Accuracy	74.44%	80.28%
Precision	17.17%	23.81%
Recall	62.96%	74.07%
F1-Score	26.98%	36.04%
AUC-ROC	0.748	0.841

On this dataset, Logistic Regression achieves the higher AUC-ROC (0.841 vs. 0.748), indicating better overall ranking of patients by risk across all decision thresholds — a useful property when the deployment team wants to tune the operating threshold post-hoc (e.g. to trade Precision for Recall as discussed in Task 2(b)).

When a statistical model is preferable:

- When the relationship between risk factors and outcome is approximately linear/log-linear (well established for many chronic-disease risk scores), Logistic Regression gives smoother, more stable probability estimates than a tree's step-function output.
- When calibrated risk probabilities are needed for downstream risk stratification (e.g. low/medium/high bands feeding into the RL treatment agent in Task 4), Logistic Regression's outputs are more directly interpretable as probabilities than Decision Tree leaf proportions.
- When the dataset is small-to-medium and there is a real risk of a tree overfitting to idiosyncratic splits, Logistic Regression's simpler hypothesis space acts as a natural regulariser.
- Conversely, the Decision Tree remains preferable when there are strong non-linear thresholds or interactions (e.g. "Glucose > 105 AND BMI > 25") that a linear model cannot capture without manual feature engineering, and when clinicians want explicit, auditable if-then rules.

3(b) Feature Importance Analysis

Rank	Decision Tree (Gini importance)	Logistic Regression (coefficient)
1	Glucose (0.4544)	Glucose (1.4581)
2	BMI (0.1828)	FamilyHistory (-1.2183)
3	BloodPressure (0.1688)	Cholesterol (0.6905)

The two models agree on **Glucose** as a shared top-3 predictor, most notably **Glucose**, which both models rank as the single most important feature — a strong, model-agnostic signal that reinforces its clinical role as the primary diagnostic marker. Where the models diverge (e.g. Logistic Regression weighting Family History more heavily than the Decision Tree does), this reflects a genuine difference in learning bias: the linear model directly captures Family History's consistent, additive effect on log-odds, while the tree — having already split on the very strong Glucose/BMI signal near the root — has less remaining impurity left for Family History to explain further down the tree. Both explanations are valid; the disagreement highlights

the value of comparing multiple inductive biases rather than trusting a single model's importance ranking in isolation.

Task 4 — Reinforcement Learning for Treatment Recommendation

4(a) MDP Formulation

Component	Definition	Justification
States	An ordinal patient health stage (Critical, HighRisk, ModerateRisk, Healthy)	A diagnosis model's risk score) gives the agent a compact, clinically meaningful summary of the patient's current health
Actions	Diet, Exercise, Medication, Move	These are the interventions a clinician can actually prescribe, ordered from least to most intensive
Reward	+10 per health stage improved; penalties for critical health events (e.g. hospitalisation)	Encourages the agent to improve patient health while avoiding inappropriate under-treatment
Transitions	Stochastic: each action results in a new health stage with a probability of improvement, or staying in the same stage	Reflects the uncertainty in the patient's response to treatment, and the stochastic nature of the environment

This is a natural fit for an MDP because (i) the Markov property holds reasonably well — the next health stage depends primarily on the current stage and the treatment applied, not the full treatment history; (ii) actions have delayed, cumulative effects (a Diet recommendation today affects health weeks later), which is exactly what the discounted return in reinforcement learning is designed to optimise; and (iii) the four treatment actions map directly onto real prescribable interventions, so the learned policy is directly actionable by clinicians.

4(b) Q-Learning Training

The agent was trained for **400 patient episodes** (exceeding the minimum of 5) using tabular Q-Learning with learning rate $\alpha = 0.1$, discount factor $\gamma = 0.9$, and an ϵ -greedy exploration policy ($\epsilon = 0.2$). The update rule applied at every step is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha \cdot [r + \gamma \cdot \max_{a'} Q(s', a') - Q(s, a)]$$

Q-table updates for 3 tracked state-action pairs across (at least) 2 iterations:

State	Action	Episode	Step	Reward	Old Q	New Q
HighRisk	Diet	1	3	10	0.0	1.0
HighRisk	Diet	1	5	0	1.0	0.99

Sample Q-table updates showing convergence of $Q(s,a)$ estimates over successive visits.

Convergence criterion: training was run until the maximum absolute change in any Q-value within an episode ($\max \Delta Q$) fell below a tolerance of 0.001, or the fixed episode budget was exhausted. The final tracked $\max \Delta Q$ was 0.11985, and Q-values were visibly stabilising across episodes (successive updates shrink geometrically, consistent with the α -weighted moving-average nature of the update rule).

4(c) Final Learned Policy & Ethical Considerations

Patient Health Stage	Recommended Action (learned policy)
Critical	Medication
HighRisk	Medication
ModerateRisk	Medication
Healthy	Diet

The learned policy recommends the more intensive **Medication** action for Critical, High-Risk and Moderate-Risk patients (states where Medication carries the highest modelled probability of improvement),

and shifts to lighter-touch **Diet/Lifestyle** maintenance once a patient reaches the Healthy state — a clinically sensible step-down in treatment intensity as health improves.

Ethical considerations for deploying an RL treatment agent:

- **Patient safety:** an RL policy trained on simulated or historical transitions can recommend actions that were never actually tested on real patients in that exact state; before deployment the policy must be validated against clinical guidelines and reviewed by a human clinician (human-in-the-loop), never auto-executed.
- **Bias:** if training data under-represents certain demographic groups (age, sex, ethnicity, socio-economic status), the learned reward-maximising policy can systematically under-treat those groups; regular fairness audits across subgroups are required before and after deployment.
- **Explainability:** unlike the Decision Tree, a Q-table/deep-RL policy does not naturally expose its reasoning; the policy should be paired with the earlier interpretable models (Task 2/3) so a clinician can see *why* a state was classified as high-risk even if the RL action-selection itself is a look-up table.
- **Reward mis-specification:** the reward function is a proxy for the true clinical goal (patient wellbeing); a poorly designed reward (e.g. purely maximising short-term "improvement" scores) can incentivise clinically inappropriate shortcuts (over-medication) unless carefully validated against long-term outcomes and side-effect costs.
- **Accountability & consent:** clear governance is needed on who is responsible for a treatment decision partly informed by an AI agent, and patients should be informed when AI-assisted recommendations are used in their care plan.

All figures referenced above (decision tree diagram, confusion matrix, ROC curves, feature-importance chart, Q-table heatmap) are provided in `diabetes_ml_output.png`, generated directly by `diabetes_ml_project.py`.